



Dual Orthogonality Sub-center Loss for Enhanced Anomalous Sound Detection

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Abstract

Anomalous Sound Detection (ASD) requires modeling a compact and discriminative normal sound distribution. Recently, angular margin loss with multiple sub-centers has been shown to be effective for ASD by extending sub-centers to capture intra-class diversity and maximizing their orthogonality to enhance model discriminability. However, existing methods do not consider that the orthogonality of intra-class sub-centers needs to be optimized based on the inherent data structure to avoid over-extension of the representation space due to over-orthogonality. To address this issue, we propose a Dual Orthogonality Sub-Center Loss (DOSCL) that enforces strict orthogonality of inter-class sub-centers to improve anomaly discrimination while applying relaxed constraints on intra-class sub-centers to capture the data structure. Experiments on the DCASE2023 Challenge Task2 dataset show that DOSCL achieves 1.74% AUC and 0.62% pAUC improvements over a strong baseline, validating its effectiveness.

Index Terms: anomalous sound detection, angular margin loss, sub-center, orthogonality

1. Introduction

Anomalous Sound Detection (ASD) is crucial for applications such as industrial monitoring [1], smart home technology [2], and health inspection [3]. However, the acquisition of anomalous sound is challenging, requiring the development of models that rely solely on normal sound data [4]. The primary goal of ASD is to create a compact and discriminative distribution of normal sounds to effectively detect anomalies that deviate from this distribution [5]. To improve the modeling of normal sound distributions for ASD, outlier exposure (OE)-based [6] methods have received considerable attention [7, 8, 9]. These methods introduce data unrelated to the target of detection as proxy anomalies, allowing the construction of classification tasks based on sample attributes [10]. The compactness and discriminative power of normal sound distributions can be further improved for OE-based methods by using angular margin losses, such as ArcFace [11], AdaCos [12]. They minimize within-class compactness while maximizing between-class separation, closely aligning with the primary objective of the ASD task [13, 14].

To more accurately capture the inherent diversity within samples, angular margin loss with multiple sub-centers, such as SCARface [15], SCAdaCos [16], have been proposed. They expand the representation space by providing multiple sub-centers for each class, thus improving the representation capability of the model [17]. These class centers are often obtained by random initialization and set to be non-trainable [16, 18], since random vectors are nearly orthogonal to each other in high-

dimensional spaces [19], which ensures discriminability between different classes.

As the number of sub-centers for each class increases from one to multiple, the relationships between these sub-centers can be categorized into two types: inter-class pairs and intra-class pairs. To optimize anomaly discrimination, inter-class pairs should be highly orthogonal [20], while intra-class pairs need to be orthogonalized by considering the underlying structure of the data to avoid representation space over-expansion [21]. However, existing methods do not consider this point and focus solely on maximizing inter-class discriminability by enforcing orthogonality among all sub-centers.

To address this issue, we propose a Dual Orthogonality Sub-center Loss (DOSCL), which categorizes sub-centers into inter-class and intra-class, imposing orthogonality constraints in a dual mechanism. DOSCL enforces strict orthogonality constraints on the inter-class cosine similarity, while applying relaxed constraints on the intra-class cosine similarity. This dual approach enables the model to enhance its ability to detect anomalies while effectively adapting to the underlying data structure. The AUC and pAUC are improved by 1.74% and 0.62%, respectively, compared to a strong baseline on the DCASE2023 Challenge Task2 dataset.

2. Background

2.1. Angular margin loss with multiple sub-centers

In traditional softmax training, the distinction between different classes is achieved by maximizing the posterior probability $\frac{e^{w_i^T x + b_i}}{\sum_{j=1}^C e^{w_j^T x + b_j}}$. Here, $x \in \mathbb{R}^d$ represents the sample embedding obtained through the feature extraction network, d is the feature dimension, W_j denotes the j -th column of the weight matrix $W \in \mathbb{R}^{d \times C}$, W is the weight of the last fully connected layer, b_j is the bias term, and C is the number of classes. However, traditional softmax only focuses on the distinction, but not the degree of distinction [22]. To address this issue, angular margin loss modifies the traditional softmax by setting $b_i = 0$ and normalizing both the weight vectors and the feature embeddings, $\|w_j\| = \|x\| = 1$ [11, 12]. Consequently, the probability of the current input belonging to class i becomes $\cos(\theta_i)$, where θ_i is the angle between the weight vector w_i and the feature x . This transformation shifts the focus from Euclidean distance to angular relationships, so that the prediction probability depends solely on the angular similarity between the feature x and the weight matrix W [23]. The key step of the angular margin loss lies in its angular margin function $f(s, \cos(\theta_i))$, which imposes an additional angular penalty on the decision margin between different classes. This approach improves the discrim-

inability by maximizing the angular separation between classes while maintaining compactness within the same class. Consequently, it improves not only the discrimination between different classes but also the overall clustering performance, resulting in more robust and discriminative feature representations for the ASD task [13].

A variety of angular margin losses have been adopted, and these can be expressed in a general form:

$$\mathcal{L}_{angular} = -\log \frac{e^{f(s, \cos(\theta_i))}}{e^{f(s, \cos(\theta_i))} + \sum_{j=1, j \neq i}^C e^{f(s, \cos(\theta_j))}} \quad (1)$$

and the $f(s, \cos(\theta_i))$ for ArcFace [11] and AdaCos [12] as follows:

$$f(s, \cos(\theta_j))_{ArcFace} = \begin{cases} s \cos(\theta_j + m) & \text{if } j = i, \\ s \cos \theta_j & \text{if } j \neq i. \end{cases} \quad (2)$$

$$f(s, \cos(\theta_j))_{AdaCos} = s(t) \cos(\theta_j) \quad (3)$$

$$s(t) = \begin{cases} \sqrt{2} \cdot \log(C-1) & \text{if } t = 0 \\ \frac{\log \hat{B}(t)_{avg}}{\cos(\min(\frac{\pi}{4}, \hat{\theta}_{med}^{(t)}))} & \text{otherwise} \end{cases} \quad (4)$$

where s is the scaling factor, m is the additional angular margin penalty, t is the number of iterations, $\hat{B}(t)_{avg}$ is the mean value computed using the angles of all the samples, and $\hat{\theta}_{med}^{(t)}$ is the median of the sample angles. Unlike the fixed angular margin penalty of Arcface, AdaCos adjusts the angular margin penalty according to the iterative process [12].

Due to the diversity of samples within a class, a single class center is insufficient for efficient representation. To address this, angular margin loss with multiple sub-centers has been introduced, e.g., SCARface [15], SCAdaCos [16]. This kind of methods expands the class-center matrix W from $d \times C \times 1$ to $d \times C \times S$, which means that the number of centers per class is extended from 1 to S . In addition, SCARface represents samples by the nearest sub-center, while SCAdaCos uses multiple sub-centers to model samples jointly, capturing more complex distributions. Therefore, the joint representation strategy is able to achieve better performance [16].

2.2. Orthogonality of class centers

As analyzed in [24], class centers should be well dispersed in the feature space to maximize inter-class differences. To achieve this, existing approaches achieve orthogonalization between class centers from different perspectives. For example, in [16], the class center matrix is initialized as an unlearnable random matrix. While this improves the orthogonality between sub-centers, it limits the ability of the model to match the underlying structure of the data. In [23], an orthogonal regularity term is imposed on the class center matrix to promote orthogonality between class centers. Other possible improvements include initializing the class center matrices as orthogonal and setting them as unlearnable. However, all of these approaches only consider the orthogonality requirement of inter-class sub-centers and cannot deal with the orthogonality requirement of sub-centers in the multiple sub-center angular margin loss, as we discuss in the next section.

3. Proposed method

3.1. Orthogonality requirements for different sub-centers

Inter-class sub-centers. To ensure that normal and proxy anomaly samples are sufficiently separated in feature space, the inter-class sub-centers should be as orthogonal as possible. This separation enhances the sensitivity of the model to anomalies. For a randomly initialized class center matrix, according to [25], the probability density function of the angle θ between sub-centers in high-dimensional space can be expressed as:

$$h(\theta) = \frac{1}{\sqrt{\pi}} \cdot \frac{\Gamma(\frac{p}{2})}{\Gamma(\frac{p-1}{2})} \cdot (\sin \theta)^{p-2}, \quad \theta \in [0, \pi] \quad (5)$$

where p denotes the dimension of sub-center, and $\Gamma(x)$ is the gamma function. In the angular margin loss framework, the dimensionality of sub-centers is determined by the feature dimensionality, which has a direct impact on the performance and efficiency of the model. Higher feature dimensionality increases the risk of over-fitting and computational cost, while lower dimensionality can lead to loss of critical information. From Eq. (5), it is known that higher dimensions generally lead to better orthogonality for randomly initialized sub-centers. However, since the dimensionality of these sub-centers is inherently matched to the feature dimensionality, insufficient feature dimensionality results in inadequate orthogonality of the randomly initialized sub-centers. On the other hand, since at most d vectors pairwise orthogonal in a d -dimensional feature, when the number of sub-centers pairs is larger than the feature dimensionality, the inter-class sub-centers should be preferentially orthogonalized to ensure the sensitivity of the model to anomalies.

Intra-class sub-centers. The size of the feature representation space for intra-class sub-centers is primarily influenced by the number of sub-centers. When there is only one class center, high-probability samples are concentrated around it, resulting in a limited representation space. As the number of sub-centers increases, samples are distributed not only around the centers but also in the intermediate regions, thereby expanding the feature space. Given a fixed number of sub-centers, the angle between these sub-centers further determines the size of the feature space [26]. However, samples in the same class should be represented in a compact space due to their inherent similarity [27], even though intra-class diversity exists. Excessive orthogonality between intra-class sub-centers can lead to an overly large and dispersed representation space. This makes the distribution of normal samples less compact and increases the risk of anomalous samples finding a suitable location within this space, reducing the effectiveness of anomaly detection.

3.2. Dual orthogonality sub-center loss (DOSCL)

In this section, we propose the dual orthogonality sub-center loss (DOSCL), a novel loss function designed to satisfy the orthogonality requirement of two types of sub-center pairs. DOSCL optimizes the orthogonality of sub-centers through a dual mechanism, which consists of two components: the inter-class orthogonality loss $\mathcal{L}_{inter}$ and the intra-class orthogonality loss $\mathcal{L}_{intra}$, as shown in Fig. 1. The DOSCL loss function is formulated as:

$$\mathcal{L}_{DOSCL} = \mathcal{L}_{inter} + \mathcal{L}_{intra} \quad (6)$$

The $\mathcal{L}_{inter}$ imposes first-order moment and second-order moment constraints on the cosine similarity between any two different classes of sub-centers (e.g., $w_c^s, w_{c'}^s$). Minimizing the

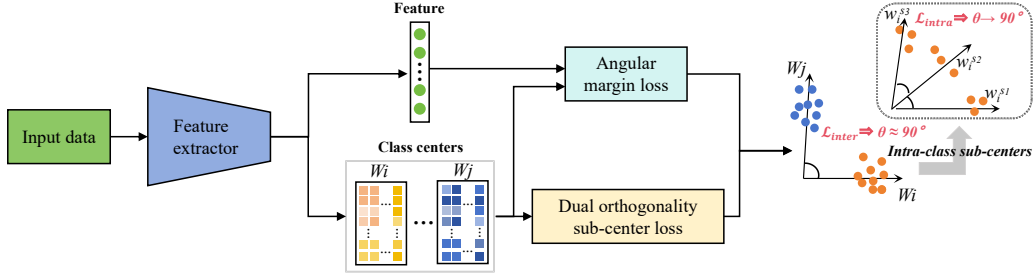


Figure 1: Schematic of the proposed DOSCL. Where W_i, W_j represent class centers of different classes, each containing multiple sub-centers (e.g. $w_i^{s1, s2, s3}$). An arbitrary sub-center constrained by both $\mathcal{L}_{inter}$ and $\mathcal{L}_{intra}$ dual orthogonality constraints.

first-order moment is equivalent to minimizing the average cosine similarity between the sub-centers, thus encouraging orthogonality between these sub-centers. Meanwhile, minimizing the second-order moment is equivalent to minimizing the variance of the cosine similarity distributions, which restricts the angle variations between sub-centers and makes the distribution more peaked, further improving orthogonality. $\mathcal{L}_{inter}$ is formulated as follows:

$$\mathcal{L}_{inter}(w_c^s, w_{c'}^{s'}) = \underbrace{\alpha \frac{1}{\binom{C}{2}} \sum_{0 \leq c < c' \leq C} \left(\frac{1}{\binom{S}{2}} \sum_{0 \leq s \leq s' \leq S} w_c^{sT} w_{c'}^{s'} \right)}_{\text{first moment}} + \underbrace{\beta \frac{1}{\binom{C}{2}} \sum_{0 \leq c < c' \leq C} \left(\frac{1}{\binom{S}{2}} \sum_{0 \leq s \leq s' \leq S} (w_c^{sT} w_{c'}^{s'})^2 \right)}_{\text{second moment}} \quad (7)$$

where C is the number of classes, c and c' are class indexes, S is the number of sub-centers for each class, s and s' are indexes of sub-centers, α and β are hyperparameters that control the influence of the first and second moments, respectively.

The $\mathcal{L}_{intra}$ minimizing the first-order moment of cosine similarity between arbitrary intra-class sub-centers (e.g., $w_c^s, w_c^{s'}$), which is equivalent to minimizing the angular means of arbitrary sub-centers within one class, which ensures that the model captures features with large intra-class differences:

$$\mathcal{L}_{intra}(w_c^s, w_c^{s'}) = \lambda \frac{1}{C} \sum_{c=1}^C \underbrace{\left(\frac{1}{\binom{S}{2}} \sum_{0 \leq s < s' \leq S} w_c^{sT} w_c^{s'} \right)}_{\text{first moment}} \quad (8)$$

In $\mathcal{L}_{intra}$, only the mean of the cosine similarity is constrained to be minimized, which allows for the existence of diverse angles between sub-centers within a class, thus matching the underlying structure of the data. This ensures that the model captures within-class variation without compromising the overall compactness of the distribution.

The hyperparameters α and λ are set equal ($\alpha = \lambda$) given the consistent physical meaning and optimization objective of the first-order moment in both inter-class and intra-class losses. This simplification thus reduces the complexity of hyperparameter tuning while maintaining theoretical consistency.

To further improve the discriminative ability of the model, the proposed DOSCL integrated with the angular margin loss

function $\mathcal{L}_{angular}$, and the overall loss function $\mathcal{L}_{all}$ is thus formulated as a combination of $\mathcal{L}_{angular}$ and $\mathcal{L}_{DOSCL}$, $\mathcal{L}_{all}$ is formulated as:

$$\mathcal{L}_{all} = \mathcal{L}_{angular} + \mathcal{L}_{DOSCL} \quad (9)$$

which enables the model to accurately and reliably differentiate between normal and anomalous samples, achieving superior performance.

4. Experimental results

4.1. Experimental setup

Experiments were conducted on the DCASE2023 Challenge Task2 dataset (“First-shot unsupervised anomalous sound detection for machine condition monitoring”) [28], which is an ASD task dataset consisting of real-world data. This dataset, sourced from MIMII DG [29] and ToyADMOS2 [30], contains fourteen device types: Fan, Gearbox, Bearing, Slide rail, Toy car, Toy train, Valve, Vacuum, ToyTank, ToyNscale, ToyDrone, bandsaw, grinder, and shaker. Each type has only one independent device data. The data from each device in dataset consist of 1000 normal samples in the training set, of which 990 are from the source domain and 10 are from the target domain, and the test set contains 200 samples, of which 100 are from the source domain and 100 are from the target domain. The samples in dataset are 10 seconds in length, with a sampling rate of 16 kHz.

Evaluation metrics including area under the receiver operating characteristic (ROC) curve (AUC) and partial AUC (pAUC) with $p = 0.1$. The metric pAUC is the AUC computed under a low false-positive rate range, namely $[0, p]$, which is used because a high true positive-rate is desirable under this conditions in practical applications to avoid frequent false alarms [31]. All experiments were conducted in the development and evaluation set of the DCASE2023 Challenge Task 2 dataset, and the results are the mean and standard deviation of five independent experiments.

The baseline was adopted from [32], an efficient ASD method that uses a finely designed model structure and is trained by SCAdaCos with randomly initialized and non-learnable class center matrix. All experiments maintain consistent network architectures and training setup with the baseline, only modifying the angular margin loss components to ensure controlled comparison. The training data is categorized into 186 classes based on the attributes of the samples. For multiple sub-center angular margin losses, each class has 16 sub-centers with feature dimensions of 512.

4.2. Performance comparison

Considering that our approach is based on angular margin loss and is related to the orthogonality of the class centers, we performed three sets of experiments: First, we evaluated the basic angular margin losses (ArcFace, AdaCos) and their trainable multiple subclass extensions (SCArcFace, SCAdaCos). Both SCArcFace and SCAdaCos utilize a joint representation strategy. Second, we investigated different strategies for the orthogonalization of sub-centers: 1) fixed centers with random initialization (*), 2) fixed centers with orthogonal initialization (†) and 3) learnable sub-centers with orthogonality loss (OL). Third, we integrated our proposed DOSCL ($\beta = 100, \alpha = 6 \times 10^3$) with existing multiple sub-center angular margin losses. The experimental results are shown in Table 1.

Table 1: Harmonic mean of AUC(%) and pAUC(%) for different methods over all machines. Highest values in bold.

Method	AUC(%)	pAUC(%)
Arcface	64.82 $\pm$ 0.39	54.63 $\pm$ 0.82
Adacos	64.91 $\pm$ 0.46	54.69 $\pm$ 0.53
SCArcface	67.22 $\pm$ 0.91	56.63 $\pm$ 0.36
SCAdaCos	67.52 $\pm$ 0.56	56.84 $\pm$ 0.70
SCArcface*	69.31 $\pm$ 0.42	57.11 $\pm$ 0.65
SCAdaCos*(baseline)	69.57 $\pm$ 0.82	57.45 $\pm$ 0.84
SCArcface†	68.40 $\pm$ 0.48	56.92 $\pm$ 0.55
SCAdaCos†	68.91 $\pm$ 0.56	57.03 $\pm$ 0.43
SCArcface+OL	69.02 $\pm$ 0.41	57.28 $\pm$ 0.63
SCAdaCos+OL	69.42 $\pm$ 0.72	57.33 $\pm$ 0.42
SCArcface+DOSCL	70.92 $\pm$ 0.75	57.81 $\pm$ 0.59
SCAdaCos+DOSCL	71.31 $\pm$ 0.61	58.07 $\pm$ 0.45

Table 1 highlights key insights on sub-center configurations. Fixed centers with orthogonal initialization (†) show less performance improvement compared to the random initialization strategy (*). In contrast, angular margin losses with learnable multiple class centers achieve improved AUC and pAUC performance when integrated with orthogonality loss (OL), compared to the non-OL configuration. However, these methods still underperform the baseline. Notably, SCAdaCos with our proposed DOSCL achieves significant performance enhancement, surpassing the baseline by 1.74% for AUC and 0.62% for pAUC.

As shown in Fig. 2, compared with fixed centers with random initialization, DOSCL achieves comparable orthogonality between inter-class sub-centers. While for the intra-class sub-centers, DOSCL results in more diverse and relatively lower orthogonality between class centers.

4.3. Performance impact of hyperparameter

The hyperparameters α and β are designed to regulate the mean and variance of sub-centers cosine similarities, respectively. A sensitivity analyzes were conducted by alternately fixing each parameter, for the SCAdaCos and SCArcFace methods with DOSCL, respectively, as shown in Fig. 3. The harmonic mean of combined AUC and pAUC $\mathcal{H}$ was chosen as the performance metric, as it provides a comprehensive measure of performance. The results show that the performance of the model depends more on the choice of α , and if α is not the optimal choice, the performance can drop significantly.

4.4. Ablation studies

To investigate the impact of different components of DOSCL on performance, ablation studies were conducted using SCAdaCos

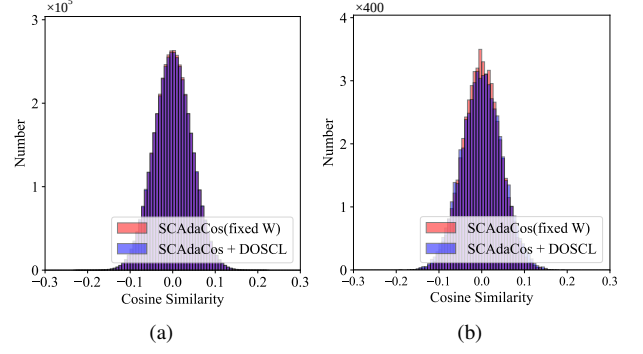


Figure 2: Histogram of cosine similarity of (a) inter-class and (b) intra-class sub-centers with fixed W and DOSCL.

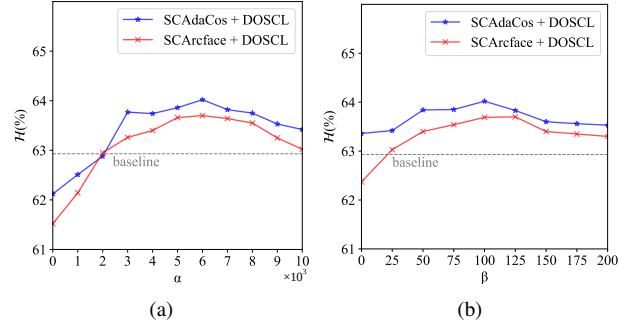


Figure 3: Harmonic mean of combined AUC and pAUC $\mathcal{H}$ (%) with varying the value of hyperparameters. (a) β fixed at 100. (b) α fixed at 6×10^3 .

as angular margin loss. The results, as shown in Table 2, indicate that the removal of any single component leads to a decline in performance. Notably, the most significant drop occurs when the first-order moment constraint of sub-centers is absent.

Table 2: Ablation studies results by using SCAdaCos as angular margin loss.

Method	AUC(%)	pAUC(%)
$\mathcal{L}_{intra}$	68.43 $\pm$ 0.56	56.87 $\pm$ 0.60
$\mathcal{L}_{inter}$ (1st moment only)	69.74 $\pm$ 0.82	57.45 $\pm$ 0.75
$\mathcal{L}_{inter}$ (2nd moment only)	68.28 $\pm$ 0.67	56.83 $\pm$ 0.56
$\mathcal{L}_{inter}$	70.46 $\pm$ 0.43	57.56 $\pm$ 0.34
$\mathcal{L}_{inter}$ (1st moment only) + $\mathcal{L}_{intra}$	70.85 $\pm$ 0.69	57.63 $\pm$ 0.83
$\mathcal{L}_{inter}$ (2nd moment only) + $\mathcal{L}_{intra}$	69.89 $\pm$ 0.94	57.47 $\pm$ 0.76

5. Conclusion

This paper analyzed the orthogonality requirements of inter-class and intra-class sub-centers within the multiple sub-center angular margin loss framework and proposed a novel DOSCL loss function to optimize their orthogonality. DOSCL ensures that inter-class sub-centers are as orthogonal as possible to enhance anomaly detection, while applying a relaxed constraint on intra-class sub-centers. This strategy allows the sub-centers to better match the underlying data structure, thereby avoiding overly constrained sub-centers that could impair the compactness of the distribution. Evaluated on the DCASE2023 Challenge Task2 dataset, DOSCL enhanced model performance compared to a strong baseline, achieving a 1.74% increase in AUC and a 0.62% increase in pAUC.

6. References

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